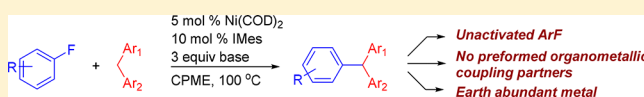


Nickel-Catalyzed C(sp³)–H Arylation of Diarylmethane Derivatives with Aryl Fluorides

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Supporting Information

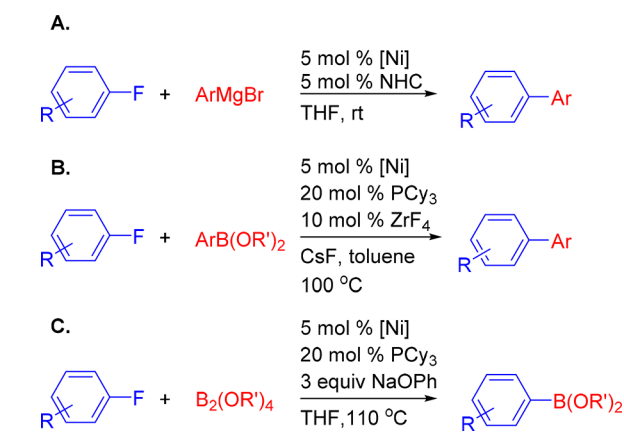
ABSTRACT: A novel nickel-catalyzed C(sp³)–H arylation with nonactivated aryl fluorides is reported. The use of 1,3-bis(2,4,6-trimethylphenyl)imidazol-2-ylidene (IMes) as a ligand was found to be critical to the success of the reaction. This new method enables the synthesis of a wide range of triarylmethane derivatives.



Transition-metal catalyzed cross-coupling reactions of aryl and alkyl halides have revolutionized the way chemists approached small molecule synthesis.¹ Compared with C–Br and C–Cl bonds, activation of C–F bonds is significantly more challenging, because of their inherent bond strength. Indeed, Ar–F bonds are among the strongest bonds in organic compounds.² Furthermore, the high electronegativity of fluorine makes the lone pairs of organofluorides poor ligands for metals. Not surprisingly, these features have made catalytic C–F bond activation an attractive and challenging goal.³ Initial success in the transition-metal-catalyzed cleavage of C–F bonds in nonactivated fluoroarenes was observed in cross-coupling reactions with Grignard^{4,5} and organozinc⁶ reagents. In pioneering work, Kumada et al. demonstrated that nickel-catalyzed cross-coupling reactions of unactivated aryl fluorides are feasible, although the yields were moderate and mixtures of products were obtained.⁷ The first truly efficient nickel-catalyzed cross-coupling reactions between aryl fluorides and aryl Grignard reagents were reported by Hermann's group using *N*-heterocyclic carbene (NHC) ligands (Scheme 1A).^{4a} Subsequently, several Kumada and Negishi cross-coupling reactions with substrates bearing C–F bonds have been reported.^{4,6} Suzuki–Miyaura cross-coupling of aryl fluorides was accomplished relatively late when Chatani and co-workers developed the first nickel-catalyzed cross-coupling reactions of monofluoroarenes with arylboronic esters. In this work a ZrF₄ co-catalyst was necessary to facilitate the C–F cleavage (Scheme 1B).^{8a} Extension of C–F activation to alkenyl fluoride substrates was recently achieved by two groups.^{8b,c} Interestingly, nickel-catalyzed borylation of aryl fluorides via C–F cleavage was independently reported by the groups of Martin and Niwa, offering new tactics for versatile functionalization of C–F bonds (Scheme 1C).⁹

Despite great advances in C–F coupling reactions, the chemistry outlined above requires the use of preformed organometallic coupling partners (e.g., RMgX and RZnX) as nucleophiles. In some cases, these reagents are very sensitive

Scheme 1. Selected Examples of Nickel-Mediated Reactions of Fluoroarenes



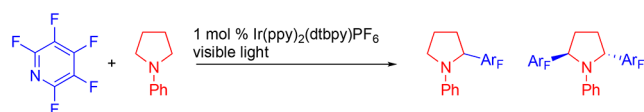
and require special handling. In other reactions, corrosive co-catalysts (ZrF₄ or TiF₄) were added to accelerate the reaction when less Lewis acidic arylboron reagents were applied as nucleophiles.^{8a} In contrast to the use of preformed organometallic reagents in couplings with aryl fluorides, less success has been achieved with pronucleophiles, wherein weakly acidic C(sp³)–H bonds are deprotonated to reveal the nucleophilic coupling partner. Very recently, Hashmi et al. developed photoredox-controlled arylation of unactivated C(sp³)–H's through C–F bond activation (Scheme 2A).¹⁰ In this chemistry, the electrophilic component was limited to highly reactive polyfluorinated benzene derivatives, which leaves substantial room for expansion. As part of our studies on deprotonative cross-coupling process (DCCP) for the arylation sp³ C–H's of diarylmethane derivatives,¹¹ we herein report the

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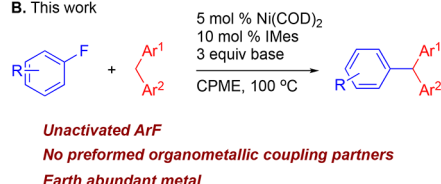
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Scheme 2. Catalytic C(sp³)-H Arylations by C-F Bond Activation

A. With reactive multifluoroarenes (Hashmi's work)



B. This work



first nickel-catalyzed DCCP reaction with monofluoroarenes using a (NHC)Ni-based catalyst (Scheme 2B).

The basic mechanism of DCCP is illustrated in Figure 1. Oxidative addition is followed by transmetalation. Before

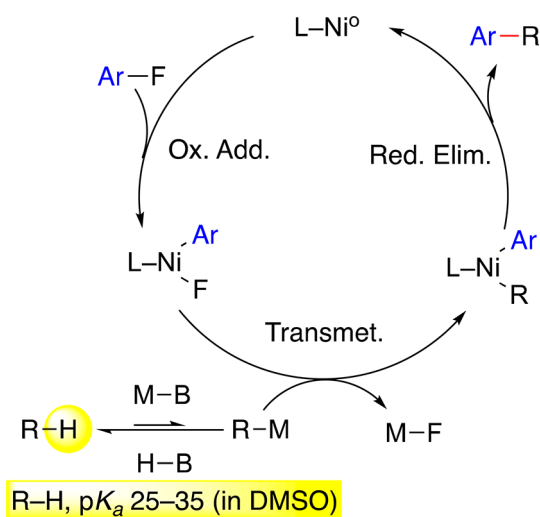


Figure 1. Generic deprotonative cross-coupling reaction mechanism. R-H is the pronucleophile, and L is a mono- or bidentate ligand.

transmetalation, the pronucleophile, which possesses weakly acidic C(sp³)-H's, undergoes reversible deprotonation. Typical pK_a values of the pronucleophiles range from 25 to 35 (in DMSO).¹² The bases are usually silylamides [MN(SiMe₃)₂, M = Li, Na, K; the pK_a of HN(SiMe₃)₂ is 26 in THF].¹³ Thus, the equilibria for deprotonation of the pronucleophiles usually lie far to the side of the substrate (R-H). After transmetalation of the nucleophile, reductive elimination takes place. The mechanism of DCCP is analogous to enolate arylations.¹⁴

We anticipated that the major challenge to achieve a successful DCCP with aryl fluorides would be to identify a catalyst capable of promoting the C-F bond activation. NIXANTPHOS (Figure 2), an excellent ligand for Pd-catalyzed DCCP with aryl bromides^{11a} and aryl chlorides,^{11c} was ineffective in this transformation. *N*-Heterocyclic carbene ligands have been successfully used with nickel to mediate C-F bond activations.^{4a,15} We, therefore, considered the use of (NHC)Ni systems for DCCP with aryl fluorides. We previously developed a complementary approach to (phosphine)Pd-catalyzed DCCP's with aryl bromides using a (NHC)Pd-based catalyst. Unfortunately, aryl chlorides and fluorides were

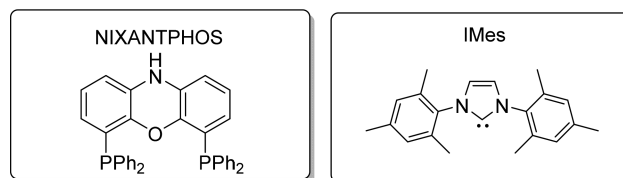


Figure 2. Structures of NIXANTPHOS and IMes.

not suitable coupling partners for this palladium-based catalyst.¹⁶

Using Ni(COD)₂ (10 mol %) and IMes (20 mol %, Figure 2) as the catalyst precursors, we examined different combinations of solvents [1,4-dioxane, DME, toluene, and cyclopentyl methyl ether (CPME)] and bases [LiO^tBu, NaO^tBu, KO^tBu, LiN(SiMe₃)₂, NaN(SiMe₃)₂, and KN(SiMe₃)₂] for the coupling reaction of 4-benzylpyridine (**1a**, 12 μmol) with 4-fluorobiphenyl (**2a**, 10 μmol) at 100 °C using high-throughput experimentation (HTE).¹⁷ Of the six bases screened, NaN(SiMe₃)₂ showed the best performance (see Supporting Information for details). On laboratory scale (0.1 mmol), four solvents (0.1 M) [1,4-dioxane, DME, toluene, and CPME] were then screened in the transformation of 4-benzylpyridine (**1a**, 1.2 equiv) and 4-fluorobiphenyl (**2a**, 1 equiv) with NaN(SiMe₃)₂ (3 equiv) using 5 mol % Ni(COD)₂/10 mol % IMes at 100 °C for 16 h. The reaction assay yields (AY) were determined by ¹H NMR integration of the unpurified reaction mixtures against an internal standard (CH₂Br₂). The results are listed in entries 1–4 (Table 1),

Table 1. Selected Optimization Results for C-H Arylation of 4-Benzylpyridine **1a**^a

entry	solvent	temp (°C)	1a:2a	assay yield ^b (%)
1	1,4-dioxane	100	1.2:1.0	12
2	DME	100	1.2:1.0	0
3	toluene	100	1.2:1.0	38
4	CPME	100	1.2:1.0	45
5	CPME	100	1.5:1.0	56
6	CPME	100	2.0:1.0	78
7	CPME	100	3.0:1.0	95 (89 ^c)
8	CPME	80	3.0:1.0	38
9	CPME	100	1.0:2.0	42

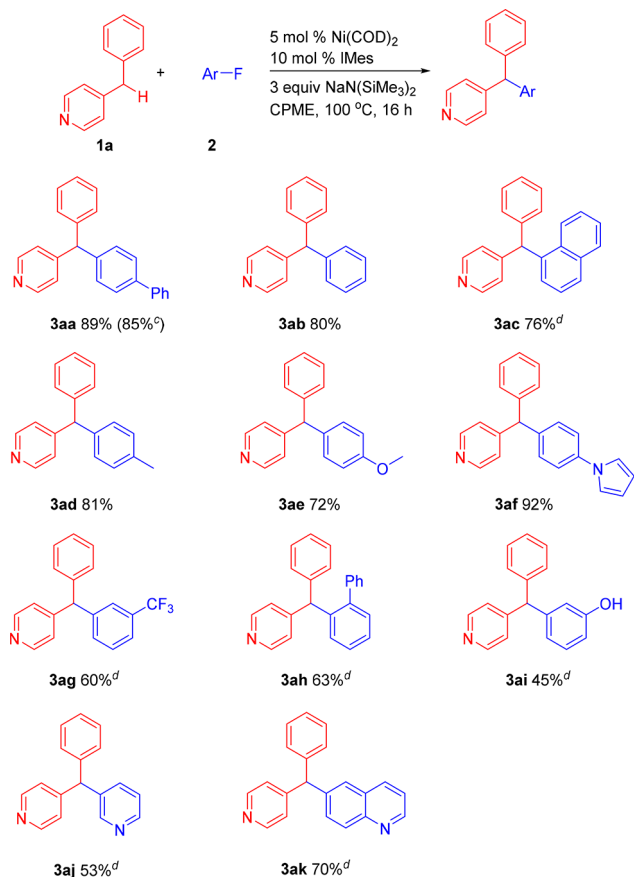
^aReactions were conducted on a 0.1 mmol scale using 1 equiv of **2a**, 3 equiv of MN(SiMe₃)₂, and **1a** at 0.1 M. ^bAssay yield determined by ¹H NMR spectroscopy of the crude reaction mixture. ^cIsolated yield after chromatographic purification.

which showed CPME gave the best result (45% AY). The ratio of 4-benzylpyridine to 4-fluorobiphenyl was next examined (entries 5–7). Fortunately, the use of 3 equiv of 4-benzylpyridine rendered 89% isolated yield (entry 7). The 100 °C reaction temperature was found to be essential for high conversion. Only 38% AY of DCCP product **3aa** was observed at 80 °C after 16 h (entry 8). Control experiments were conducted in the absence of ligand or nickel and resulted in no product. The excess amount of 4-benzylpyridine (total 3 equiv) is critical in this transformation. When 2 equiv of 4-

fluorobiphenyl was used, the coupling product was obtained in only 42% yield (entry 9), with no tetraarylmethane observed. The optimized conditions (entry 7, Table 1) were then carried forward to define the substrate scope.

The scope of the DCCP with 4-benzylpyridine (**1a**) and various aryl fluorides is presented in Table 2. All reactions were

Table 2. Scope of Aryl Fluorides in Ni-Catalyzed DCCP^{a,b}



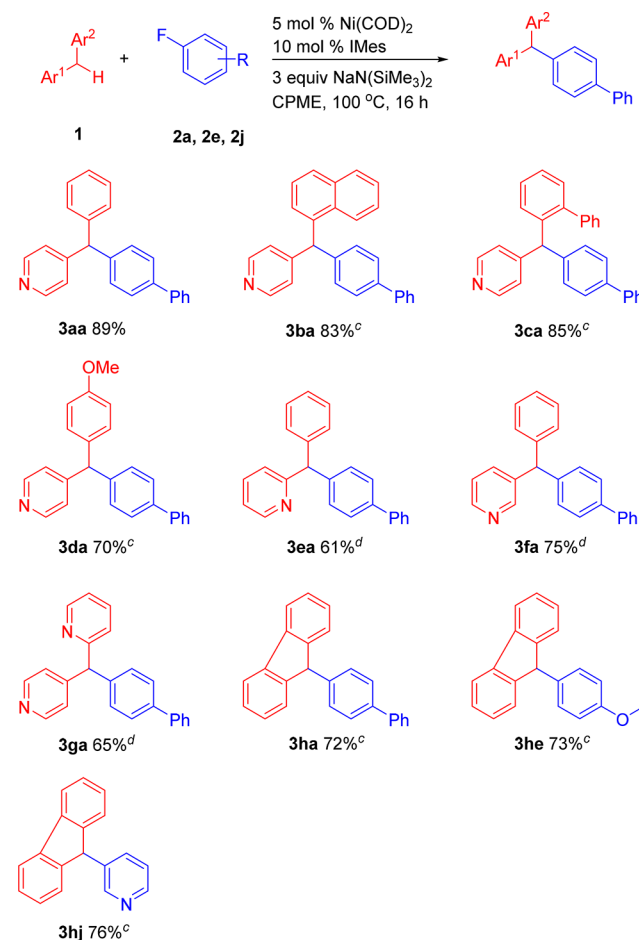
^aReactions were conducted on a 0.1 mmol scale using 1 equiv of aryl fluoride, 3 equiv of NaN(SiMe₃)₂, and 3 equiv of **1a** at 0.1 M. ^bIsolated yield after chromatographic purification. ^cReaction conducted on 5 mmol scale. ^d120 °C.

performed at 100 °C, except where noted. Phenyl and 1-naphthyl fluorides provided **3ab** and **3ac** in 80 and 76% yield, respectively. Aryl fluorides bearing different substituents exhibited good to excellent reactivity, including those with alkyl (**3ad**), electron-donating (**3ae**, **3af**), electron-withdrawing (**3ag**), and *ortho* substituents (**3ah**). In addition, an aryl fluoride bearing a hydroxyl group (3-fluorophenol) was tolerated in this reaction giving 45% yield of **3ai** with 3 equiv of base at higher temperature. In contrast, 4-fluorobenzamide decomposed under the reaction conditions. Heterocyclic 3-fluoropyridine and 6-fluoroquinoline were also suitable coupling partners, furnishing products **3aj**–**3ak** in 53–70% yields. We tested heterocyclic substrates such as 5-fluorobenzofuran and 5-fluorobenzothiophene. Unfortunately, these substrates gave multiple products.

To illustrate the scalability of this method, we conducted the coupling reaction of 4-benzylpyridine (**1a**) and 4-fluorobiphenyl (**2a**) on a 5 mmol scale. The DCCP product **3aa** was isolated in 85% yield (1.36 g).

The scope of diarylmethanes in the DCCP was next explored with 4-fluorobiphenyl (**2a**) (Table 3). Upon substituting the

Table 3. Scope of Diarylmethanes in Ni-Catalyzed DCCP^{a,b}



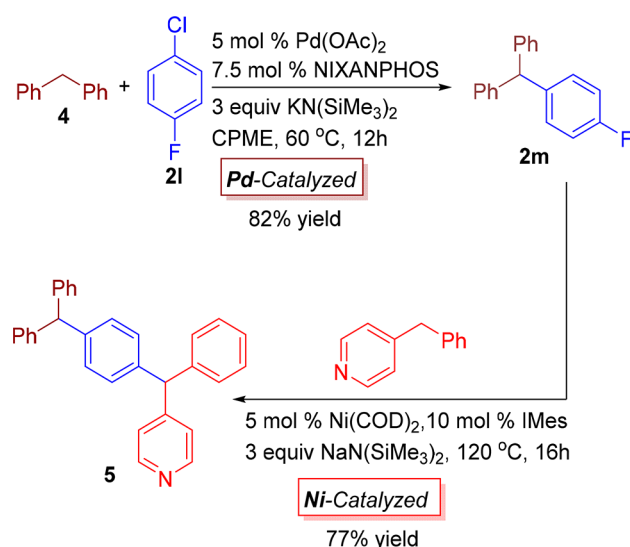
^aReactions were conducted on a 0.1 mmol scale using 1 equiv of **2**, 3 equiv of NaN(SiMe₃)₂, and 3 equiv of **1** at 0.1 M. ^bIsolated yield after chromatographic purification. ^c120 °C. ^d3 equiv of KN(SiMe₃)₂ and 6 equiv of 18-crown-6.

phenyl group in 4-benzylpyridine with 1-naphthyl (**2b**) or 2-biphenyl (**2c**), the catalyst was demonstrated to be tolerant of sterically demanding substrates (83 and 85% yield, respectively). Use of a substrate bearing a 4-methoxyphenyl group (**2d**), which renders the benzylic hydrogens more difficult to deprotonate, resulted in a slightly diminished yield of the product **3da** (70%). Likewise, the isomeric 2- and 3-benzylpyridine derivatives, which are also less acidic than the parent 4-benzylpyridine,^{12,18} exhibited decreased reactivity under our standard conditions. We previously demonstrated that the addition of crown ethers can dramatically increase the reactivity in DCCP's by reducing the aggregation state of the base.^{10b} As illustrated in Table 3, when 3 equiv KN(SiMe₃)₂/6 equiv 18-crown-6 was used to replace NaN(SiMe₃)₂, 2-benzylpyridine (**1e**) and 3-benzylpyridine (**1f**) performed well in the DCCP, generating the coupling products in 61 and 75% yield, respectively. Furthermore, the more acidic 2,4'-dipyridinylmethane (**1g**) underwent coupling with aryl fluoride **2a** giving the product **3ga** in 65% yield. It is noteworthy that pyridyl groups are among the most common heterocycles in bioactive compounds.¹⁹ For diarylmethane derivatives void of

heteroaryl groups, fluorene was examined. Use of fluorene with 4-fluorobiphenyl, 4-fluoroanisole, and 3-fluoropyridine at 120 °C provided coupling products **3ha**, **3he**, and **3hj** in 72–76% yields. Diphenylmethane was also tested under these conditions, however, the starting materials were recovered, even when 3 equiv $\text{KN}(\text{SiMe}_3)_2$ was used with or without 6 equiv 18-crown-6. This result may be due to the higher pK_a of diphenylmethane relative to the substrates successfully employed in Table 3.¹²

The robust nature of the C–F bond allows the fluorine to withstand nearly all reactions in organic chemistry, enabling the aryl fluoride coupling reaction to be conducted at a late stage in the synthesis. To showcase the advantages of the aryl fluoride coupling, sequential palladium- and nickel-catalyzed DCCP were performed (Scheme 3). Thus, 1-chloro-4-fluorobenzene

Scheme 3. Sequential Pd- and Ni-Catalyzed DCCP



(**21**) was first cross-coupled with diphenylmethane (**4**) using a (NIXANTPHOS)Pd catalyst, providing the coupling product **2m** in 82% yield.^{11c} No C–F coupling product was observed. Next, **2m** underwent nickel-catalyzed DCCP with 4-benzylpyridine **1a** to provide the desired product **5** in 77% yield.

In summary, we have developed the first nickel-catalyzed DCCP with unactivated C–F bonds of aryl fluorides. The use of IMes as a ligand was found to be critical to the success of this transformation, because of its strong σ -donor ability,²⁰ which facilitates the oxidative addition step of the catalytic cycle. This new protocol is distinct from our previous DCCP, because it involves the cleavage of an inert C–F bond. Based on this unique feature, we envision that this method will find applications in organic synthesis, especially given that many aryl fluorides are commercially available and nickel is earth abundant and excellent in coupling reactions.²¹ Further studies toward the development of new catalytic methods involving the activation of C–F bonds are currently underway in our laboratories.

EXPERIMENTAL SECTION

General Methods. All reactions were conducted under an atmosphere of dry nitrogen. Anhydrous cyclopentyl methyl ether (CPME), dioxane, toluene, and dimethoxyethane (DME) were purchased from Sigma-Aldrich and used without further purification. Unless otherwise stated, all reagents were commercially available and

used as received without further purification. Flash chromatography was performed with silica gel (300–400 mesh). The infrared spectra were obtained with KBr plates using a PerkinElmer Spectrum 1600 Series spectrometer. NMR spectra were obtained using a Bruker AM-500 Fourier-transform NMR spectrometer at 500 and 125 MHz, respectively. Chemical shifts are reported in units of parts per million (ppm) downfield from tetramethylsilane (TMS), and all coupling constants are reported in hertz. High-resolution mass spectrometry (HRMS) data were obtained on a Waters LC-TOF mass spectrometer (model LCT-XE Premier) using chemical ionization (CI) or electrospray ionization (ESI) in positive or negative mode, depending on the analyte. Melting points were determined on a Unimelt Thomas-Hoover melting point apparatus and were uncorrected.

Procedure and Characterization for Ni-Catalyzed DCCP of Diarylmethanes. *General Procedure A.* An oven-dried 8 mL vial equipped with a stir bar was charged with $\text{NaN}(\text{SiMe}_3)_2$ (55 mg, 0.3 mmol, 3.0 equiv) under a nitrogen atmosphere in the glovebox. A solution (from a stock solution) of $\text{Ni}(\text{COD})_2$ (1.4 mg, 0.005 mmol) and 1,3-bis(2,4,6-trimethylphenyl)imidazol-2-ylidene (IMes) (3.0 mg, 0.01 mmol) in 1.0 mL of dry CPME was taken up by syringe and added to the mixture. After stirring for 10 min at 24 °C, diarylmethane (0.3 mmol, 3.0 equiv) was added to the reaction mixture, followed by aryl fluoride (0.1 mmol). The vial was capped, removed from the glovebox, and stirred for 16 h at 100 °C (or 120 °C). After cooling to room temperature, the reaction mixture was quenched with H_2O (three drops) and passed through a short pad of silica gel and eluted with ethyl acetate (1 mL $\times$ 3). The combined organics were concentrated under reduced pressure. The crude material was loaded onto a silica gel column and purified by flash chromatography.

General Procedure B. An oven-dried 8 mL vial equipped with a stir bar in a nitrogen filled drybox was charged with $\text{KN}(\text{SiMe}_3)_2$ (59.8 mg, 0.3 mmol, 3.0 equiv) and 18-crown-6 (158.6 mg, 0.6 mmol, 6.0 equiv). A solution (from a stock solution) of $\text{Ni}(\text{COD})_2$ (1.4 mg, 0.005 mmol) and 1,3-bis(2,4,6-trimethylphenyl)imidazol-2-ylidene (IMes) (3.0 mg, 0.01 mmol) in 1.0 mL of dry CPME was taken up by syringe and added to the mixture. After stirring for 10 min at room temperature, diarylmethane (0.3 mmol, 3.0 equiv) was added to the reaction mixture, followed by aryl fluoride (0.1 mmol). The vial was capped, removed from the glovebox, and stirred for 16 h at 100 °C. After cooling to room temperature, the reaction mixture was quenched with H_2O (three drops) and passed through a short pad of silica gel and eluted with ethyl acetate (1 mL $\times$ 3). The combined organics were concentrated in vacuo. The crude material was loaded onto a silica gel column and purified by flash chromatography.

4-([1,1'-Biphenyl]-4-yl(phenyl)methyl)pyridine (**3aa**).¹⁶ The reaction was performed following General Procedure A with **1a** (47.8 μL , 0.3 mmol), $\text{NaN}(\text{SiMe}_3)_2$ (55 mg, 0.3 mmol), and **2a** (17.2 mg, 0.1 mmol) at 100 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (28.6 mg, 89% yield) as a white solid. The NMR spectral data match the previously published data.

Diphenyl(4-pyridyl)methane (**3ab**).²² The reaction was performed following General Procedure A with **1a** (47.8 μL , 0.3 mmol), $\text{NaN}(\text{SiMe}_3)_2$ (55 mg, 0.3 mmol), and fluorobenzene (**2b**) (9.3 μL , 0.1 mmol) at 100 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (19.6 mg, 80% yield) as a white solid. The NMR spectral data match the previously published data.

4-(Naphthalen-1-yl(phenyl)methyl)pyridine (**3ac**).¹⁶ The reaction was performed following General Procedure A with **1a** (47.8 μL , 0.3 mmol), $\text{NaN}(\text{SiMe}_3)_2$ (55 mg, 0.3 mmol), and 1-fluoronaphthalene (**2c**) (12.9 μL , 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (22.4 mg, 76% yield) as a white solid. The NMR spectral data match the previously published data.

4-(Phenyl(*p*-tolyl)methyl)pyridine (**3ad**).²³ The reaction was performed following General Procedure A with **1a** (47.8 μL , 0.3 mmol), $\text{NaN}(\text{SiMe}_3)_2$ (55 mg, 0.3 mmol), and 4-fluorotoluene (**2d**) (11 μL , 0.1 mmol) at 100 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to

give the product (21 mg, 81% yield) as a white solid. The NMR spectral data match the previously published data.

(4-Methoxyphenyl)(4-pyridyl)phenylmethane (3ae).²² The reaction was performed following General Procedure A with **1a** (47.8 μ L, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and 4-fluoroanisole (**2e**) (11.3 μ L, 0.1 mmol) at 100 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (22.0 mg, 80% yield) as a colorless oil. The NMR spectral data match the previously published data.

4-((4-(1H-Pyrrol-1-yl)phenyl)(phenyl)methyl)pyridine (3af). The reaction was performed following General Procedure A with **1a** (47.8 μ L, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and 1-(4-fluorophenyl)pyrrole (**2f**) (16.1 mg, 0.1 mmol) at 100 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 6:1) to give the product (28.5 mg, 92% yield) as a white solid. Mp = 84–85 °C; ¹H NMR (500 MHz, CDCl₃): δ 8.55–8.53 (m, 2H), 7.35–7.32 (m, 4H), 7.29–7.26 (m, 1H), 7.16–7.11 (m, 4H), 7.07–7.06 (m, 4H), 6.34 (t, J = 2.2 Hz, 2H) 5.52 (s, 1H); ¹³C{¹H} NMR (125 MHz, CDCl₃): δ 152.5, 150.0, 141.9, 139.5, 130.4, 129.3, 128.7, 127.1, 124.6, 120.6, 119.2, 110.5, 55.7, one resonance was not observed due to coincidental overlap; IR (thin film): 2923, 1593, 1488, 1411 cm⁻¹; HRMS (ESI) m/z : [M + H]⁺ calcd for C₂₂H₁₉N₂ 311.1548; found 311.1576.

(3-Trifluoromethylphenyl)(4-pyridyl)phenylmethane (3ag). The reaction was performed following General Procedure A with **1a** (47.8 μ L, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and 3-fluorobenzotrifluoride (**2g**) (16.4 mg, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (18.6 mg, 60% yield) as a colorless oil. ¹H NMR (500 MHz, CDCl₃): δ 8.84–8.34 (m, 2H), 7.53 (d, J = 7.7 Hz, 1H), 7.44 (t, J = 7.8 Hz, 1H), 7.38 (s, 1H), 7.34 (t, J = 7.5 Hz, 2H), 7.31–7.27 (m, 2H), 7.08 (d, J = 7.6 Hz, 2H), 7.04–7.00 (m, 2H), 5.55 (s, 1H); ¹³C{¹H} NMR (125 MHz, CDCl₃): δ 151.9, 150.2, 143.3, 141.3, 132.8, 131.2, 130.9, 129.3, 129.2, 128.9, 127.4, 126.1 (q, J = 3.7 Hz), 124.5, 123.9 (q, J = 3.7 Hz), 56.14; IR (thin film): 3027, 2923, 1594, 1493, 1447, 1414 cm⁻¹; HRMS (ESI) m/z : [M + H]⁺ calcd for C₁₉H₁₅F₃N 314.1157; found 314.1175.

4-((1,1'-Biphenyl)-2-yl(phenyl)methyl)pyridine (3ah).¹⁶ The reaction was performed following General Procedure A with **1a** (47.8 μ L, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and 2-fluorobiphenyl (**2h**) (17.2 mg, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (20.2 mg, 63% yield) as a white solid. The NMR spectral data match the previously published data.

3-(Phenyl(pyridin-4-yl)methyl)phenol (3ai). The reaction was performed following General Procedure A with **1a** (47.8 μ L, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and 3-fluorophenol (**2i**) (11.2 mg, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (11.7 mg, 45% yield) as a white solid. Mp = 95–98 °C; ¹H NMR (500 MHz, CDCl₃): δ 9.26 (s, 1H), 8.32 (s, 2H), 7.31 (t, J = 7.6 Hz, 2H), 7.25 (d, J = 4.7 Hz, 2H), 7.19 (t, J = 7.9 Hz, 1H), 7.09 (d, J = 7.5 Hz, 2H), 7.05 (d, J = 6.0 Hz, 2H), 6.78 (dd, J = 8.1, 2.4 Hz, 1H), 6.67 (d, J = 7.6 Hz, 1H), 6.43 (s, 1H), 5.46 (s, 1H); ¹³C{¹H} NMR (125 MHz, CDCl₃): δ 158.0, 154.4, 148.9, 143.8, 142.3, 130.4, 129.8, 129.1, 127.5, 125.6, 121.6, 116.6, 114.9, 56.4; IR (thin film): 2922, 1597, 1493, 1454, 1419, 1267 cm⁻¹; HRMS (ESI) m/z : [M]⁺ calcd for C₁₈H₁₅NO 261.1154; found 261.1159.

3-(Phenyl(pyridin-4-yl)methyl)pyridine (3aj). The reaction was performed following General Procedure A with **1a** (47.8 μ L, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and 3-fluoropyridine (**2j**) (8.6 μ L, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with EtOAc) to give the product (15.5 mg, 63% yield) as a colorless oil. ¹H NMR (500 MHz, CDCl₃): δ 8.53–8.51 (m, 2H), 8.49 (dd, J = 4.8, 1.6 Hz, 1H), 8.41 (d, J = 2.3 Hz, 1H) 7.38–7.21 (m, 5H), 7.08–7.06 (m, 2H), 7.02–7.00 (m, 2H), 5.49 (s, 1H); ¹³C{¹H} NMR (125 MHz, CDCl₃): δ 151.5, 150.6, 150.0, 148.3, 140.8, 137.6, 136.5, 129.1, 128.8, 127.2, 124.3, 123.4, 53.7; IR (thin film): 3395, 3029, 1595, 1493, 1416 cm⁻¹; HRMS (ESI) m/z : [M + H]⁺ calcd for C₁₇H₁₅N₂ 247.1235; found 247.1254.

6-(Phenyl(pyridin-4-yl)methyl)quinoline (3ak). The reaction was performed following General Procedure A with **1a** (47.8 μ L, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and 6-fluoroquinoline (**2k**) (12.2 μ L, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with EtOAc) to give the product (20.7 mg, 70% yield) as a colorless oil. ¹H NMR (500 MHz, CDCl₃): δ 8.88 (dd, J = 4.2, 1.7 Hz, 1H), 8.53 (d, J = 6.0 Hz, 2H), 8.03 (d, J = 8.5 Hz, 2H), 7.49 (dd, J = 8.7, 2.0 Hz, 1H), 7.41–7.24 (m, 5H), 7.13–7.11 (m, 2H), 7.07–7.06 (m, 2H), 5.67 (s, 1H); ¹³C{¹H} NMR (125 MHz, CDCl₃): δ 152.1, 150.5, 150.0, 147.4, 141.5, 140.6, 136.0, 131.2, 129.8, 129.4, 128.8, 128.2, 127.7, 127.2, 124.7, 121.4, 56.2; IR (thin film): 3027, 2923, 1592, 1493, 1414 cm⁻¹; HRMS (ESI) m/z : [M + H]⁺ calcd for C₂₁H₁₇N₂ 297.1392; found 297.1387.

4-((1,1'-Biphenyl)-4-yl(naphthalen-1-yl)methyl)pyridine (3ba). The reaction was performed following General Procedure A with 4-(naphthalen-1-ylmethyl)pyridine (**1b**) (65.9 mg, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and **2a** (17.2 mg, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 6:1) to give the product (30.8 mg, 83% yield) as a colorless oil. ¹H NMR (500 MHz, CDCl₃): δ 8.53–8.52 (m, 2H), 7.92 (d, J = 8.3 Hz, 1H), 7.88 (d, J = 8.0 Hz, 1H), 7.80 (d, J = 8.2 Hz, 1H), 7.58–7.31 (m, 10H), 7.16 (d, J = 8.1 Hz, 2H), 7.08–7.07 (m, 2H), 6.97 (d, J = 7.2 Hz, 1H), 6.26 (s, 1H); ¹³C{¹H} NMR (125 MHz, CDCl₃): δ 152.9, 149.9, 140.9, 140.5, 139.9, 138.0, 134.0, 131.7, 130.0, 128.9, 128.8, 128.0, 127.6, 127.4, 127.0, 126.5, 125.8, 125.3, 125.0, 123.9, 52.3, one resonance was not observed due to coincidental overlap; IR (thin film): 3051, 2923, 2853, 1596, 1488, 1411 cm⁻¹; HRMS (ESI) m/z : [M + H]⁺ calcd for C₂₈H₂₂N 372.1752; found 372.1730.

4-((1,1'-Biphenyl)-2-yl((1,1'-biphenyl)-4-yl)methyl)pyridine (3ca). The reaction was performed following General Procedure A with 2-fluoro-1,1'-biphenyl (**1c**) (73.5 mg, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and **2a** (17.2 mg, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 6:1) to give the product (33.7 mg, 85% yield) as a colorless oil. ¹H NMR (500 MHz, CDCl₃): δ 8.58–8.56 (m, 2H), 7.61–7.52 (m, 7H), 7.46–7.33 (m, 8H), 7.23 (d, J = 8.2 Hz, 2H), 7.15–7.13 (m, 3H), 5.62 (s, 1H); ¹³C{¹H} NMR (125 MHz, CDCl₃): δ 152.7, 150.0, 142.6, 141.7, 141.1, 140.9, 140.6, 139.9, 129.8, 129.1, 128.8, 128.3, 128.2, 127.5, 127.40, 127.38, 127.2, 127.1, 125.9, 124.7, 56.1, one resonance was not observed due to coincidental overlap; IR (thin film): 3029, 2922, 1595, 1484, 1413 cm⁻¹; HRMS (ESI) m/z : [M + H]⁺ calcd for C₃₀H₂₄N 398.1909; found 398.1881.

4-((1,1'-Biphenyl)-4-yl(4-methoxyphenyl)methyl)pyridine (3da). The reaction was performed following General Procedure A with 4-(4-methoxybenzyl)pyridine (**1d**) (59.7 mg, 0.3 mmol), NaN(SiMe₃)₂ (55 mg, 0.3 mmol), and **2a** (17.2 mg, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (24.6 mg, 70% yield) as a colorless oil. ¹H NMR (500 MHz, CDCl₃): δ 8.52–8.51 (m, 2H), 7.57–7.52 (m, 4H), 7.43–7.31 (m, 3H), 7.15 (d, J = 8.1 Hz, 2H), 7.07 (d, J = 6.0 Hz, 2H), 7.04 (d, J = 8.7 Hz, 2H) 6.86–6.85 (m, 2H), 5.47 (s, 1H), 3.79 (s, 3H); ¹³C{¹H} NMR (125 MHz, CDCl₃): δ 158.4, 153.0, 149.8, 141.5, 140.5, 139.6, 134.1, 130.2, 129.6, 128.7, 127.3, 127.2, 127.0, 124.5, 114.0, 55.2, 55.1; IR (thin film): 3029, 3285, 2851, 1596, 1509, 1412 cm⁻¹; HRMS (ESI) m/z : [M + H]⁺ calcd for C₂₅H₂₂NO 352.1701; found 352.1696.

2-((1,1'-Biphenyl)-4-yl(phenyl)methyl)pyridine (3ea). The reaction was performed following General Procedure B with 2-benzylpyridine (**1e**) (50.7 mg, 0.3 mmol), KN(SiMe₃)₂ (59.9 mg, 0.3 mmol), 18-crown-6 (158.6 mg, 0.6 mmol), and **2a** (17.2 mg, 0.1 mmol) at 100 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 8:1) to give the product (19.6 mg, 61% yield) as a white solid. Mp = 68–69 °C; ¹H NMR (500 MHz, CDCl₃): δ 8.64–8.63 (m, 1H), 7.65–7.52 (m, 5H), 7.44–7.41 (m, 2H), 7.34–7.31 (m, 3H), 7.25–7.22 (m, 5H), 7.17–7.14 (m, 2H), 5.75 (s, 1H); ¹³C{¹H} NMR (125 MHz, CDCl₃): δ 163.2, 149.7, 142.7, 141.9, 140.9, 139.4, 136.5, 129.8, 129.4, 128.7, 128.5, 127.18, 127.17, 127.1, 126.6, 123.8, 121.5, 59.1; IR (thin film): 3027, 2964,

2923, 1586, 1486, 1432 cm^{-1} ; HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{24}\text{H}_{20}\text{N}$ 322.1596; found 322.1617.

3-((1,1'-Biphenyl)-4-yl(phenyl)methyl)pyridine (3fa). The reaction was performed following General Procedure B with 3-benzylpyridine (1f) (50.7 mg, 0.3 mmol), $\text{KN}(\text{SiMe}_3)_2$ (59.9 mg, 0.3 mmol), 18-crown-6 (158.6 mg, 0.6 mmol), and **2a** (17.2 mg, 0.1 mmol) at 100 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 8:1) to give the product (24.1 mg, 75% yield) as a white solid. Mp = 75–76 °C; ^1H NMR (500 MHz, CDCl_3): δ 8.50–8.48 (m, 2H), 7.59–7.53 (m, 4H), 7.46–7.42 (m, 3H), 7.36–7.32 (m, 3H), 7.28–7.23 (m, 2H), 7.19–7.15 (m, 4H), 5.60 (s, 1H); $^{13}\text{C}\{^1\text{H}\}$ NMR (125 MHz, CDCl_3): δ 150.8, 147.8, 142.6, 141.7, 140.6, 139.6, 139.2, 136.6, 129.6, 129.2, 128.7, 128.6, 127.23, 127.20, 127.0, 126.7, 123.2, 54.0; IR (thin film): 3026, 2922, 1575, 1486, 1450 cm^{-1} ; HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{24}\text{H}_{20}\text{N}$ 322.1596; found 322.1605.

2-((1,1'-Biphenyl)-4-yl(pyridin-4-yl)methyl)pyridine (3ga). The reaction was performed following General Procedure B with 2-(pyridin-4-ylmethyl)pyridine (1g) (51 mg, 0.3 mmol), $\text{KN}(\text{SiMe}_3)_2$ (59.9 mg, 0.3 mmol), 18-crown-6 (158.6 mg, 0.6 mmol), and **2a** (17.2 mg, 0.1 mmol) at 100 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 1:3) to give the product (20.9 mg, 65% yield) as a white solid. Mp = 76–77 °C; ^1H NMR (500 MHz, CDCl_3): δ 8.64–8.63 (m, 1H), 8.55–8.54 (m, 2H), 7.67–7.64 (m, 1H), 7.58–7.55 (m, 4H), 7.44–7.41 (m, 2H), 7.35–7.32 (m, 1H), 7.26–7.25 (m, 2H), 7.21–7.18 (m, 1H), 7.16–7.14 (m, 3H), 5.67 (s, 1H); $^{13}\text{C}\{^1\text{H}\}$ NMR (125 MHz, CDCl_3): δ 161.4, 151.6, 149.94, 149.88, 140.6, 140.1, 140.0, 136.8, 129.7, 128.8, 127.44, 127.38, 127.1, 124.6, 123.8, 122.0, 58.3; IR (thin film): 3402, 2922, 2851, 1586, 1485, 1432 cm^{-1} ; HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{23}\text{H}_{19}\text{N}_2$ 323.1548; found 323.1552.

9-((1,1'-Biphenyl)-4-yl)-9H-fluorene (3ha).²⁴ The reaction was performed following General Procedure A with fluorene (1h) (49.8 mg, 0.3 mmol), $\text{NaN}(\text{SiMe}_3)_2$ (55 mg, 0.3 mmol), and **2a** (17.2 mg, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes) to give the product (22.9 mg, 72% yield) as a white solid. The NMR spectral data match the previously published data.

9-(4-Methoxyphenyl)-9H-fluorene (3he).²⁴ The reaction was performed following General Procedure A with fluorene (1h) (49.8 mg, 0.3 mmol), $\text{NaN}(\text{SiMe}_3)_2$ (55 mg, 0.3 mmol), and **2e** (11.3 μL , 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (19.8 mg, 73% yield) as a white solid. The NMR spectral data match the previously published data.

3-(9H-Fluoren-9-yl)pyridine (3hj).²⁵ The reaction was performed following General Procedure A with fluorene (1h) (49.8 mg, 0.3 mmol), $\text{NaN}(\text{SiMe}_3)_2$ (55 mg, 0.3 mmol), and **2j** (8.6 μL , 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 5:1) to give the product (18.5 mg, 76% yield) as a pale yellow solid. The NMR spectral data match the previously published data.

((4-Fluorophenyl)methylene)dibenzene (2m).^{11c} The compound was prepared according to literature procedures with diphenylmethane (4) (33.4 μL , 0.2 mmol), $\text{KN}(\text{SiMe}_3)_2$ (119.6 mg, 0.6 mmol), and 1-chloro-4-fluorobenzene (2l) (42.6 μL , 0.4 mmol) at 60 °C.⁵ The crude material was purified by flash chromatography on silica gel (eluted with hexanes) to give the product (43.0 mg, 82% yield) as a white solid. The NMR spectral data match the previously published data.

4-((4-Benzhydrylphenyl)(phenyl)methyl)pyridine (5). The reaction was performed following General Procedure A with **1a** (50.7 mg, 0.3 mmol), $\text{NaN}(\text{SiMe}_3)_2$ (55 mg, 0.3 mmol), and **2m** (26.2 mg, 0.1 mmol) at 120 °C. The crude material was purified by flash chromatography on silica gel (eluted with hexanes:EtOAc = 8:1) to give the product (31.7 mg, 77% yield) as a white solid. Mp = 125–126 °C; ^1H NMR (500 MHz, CDCl_3): δ 8.51–8.49 (d, J = 5.2 Hz, 2H), 7.30–7.18 (m, 10H), 7.11–6.98 (m, 11H), 5.50 (s, 1H), 5.44 (s, 1H); $^{13}\text{C}\{^1\text{H}\}$ NMR (125 MHz, CDCl_3): δ 152.7, 149.7, 143.7, 142.4, 142.1, 139.9, 129.5, 129.33, 129.25, 129.1, 128.5, 128.3, 126.8, 126.3, 124.5, 56.4, 55.8; IR (thin film): 3394, 3024, 2920, 2851, 1595, 1493,

1450 cm^{-1} ; HRMS (ESI) m/z : $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{31}\text{H}_{26}\text{N}$ 412.2065; found 412.2036.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.joc.8b00016.

Screening procedures, characterization data for all new compounds (PDF)

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Notes

The authors declare no competing financial interest.

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